NEVO / ENGINEERING REVIEW

# Learning science: the build and the gaps

**Backend audit and implementation plan | 18 October 2026**

**Bottom line:** Nevo has substantial adaptive delivery infrastructure, but the supplied learning-science document is not yet a literal description of the complete implementation. The largest differences are measurement reduction, daily recalibration, the named knowledge-tracing models, and curriculum graph coverage.

This report identifies **12 backend gaps or contract risks**, one cross-layer affective-inference issue, and one explicit product-policy conflict. It does not claim that every endpoint is broken, nor that passing tests proves educational effectiveness.

## Scope and evidence

All 12 pages of *Nevo_Learning_Science.md.pdf* were reviewed against backend commit **1dcf126**, relevant repository browser-helper code, existing checks, read-only database aggregates and selected live API reads. No frontend application code was changed or browser behaviour certified. Provider credits are outside this audit.

## Rollout issue found and resolved

The database initially reported 0104 despite the deployed API advertising the new flag routes. The live E2E teacher flags read returned 500 (incident **f2bbc206d9e4**). At your request, migration **20261018_0105** was applied successfully. The database now reports 0105, and the same live flags read returns **200**. This is resolved, not an open learning-science gap.

## Verification boundaries

54 selected existing checks passed. Additional controlled function checks confirmed specific gaps described below. Existing tests validate today's rules; they do not establish that those rules implement BKT, IRT or deep AKT. No new test files or application fixes were added during this audit.

Priority key: P1 = correctness or delivery work to address first; P2 = model/claim alignment requiring an explicit product and engineering decision. Proposed fixes below are not shipped changes.

COVERAGE MAP

# What is already there

| Mechanism / source pages | Backend position | Qualification |
|---|---|---|
| Four-module onboarding / pp. 3-5 | Implemented with exception | Gate, resume and legacy credit exist; no-bank waiver differs from the document (D1). |
| Motor correction / p. 4 | Implemented, conditional | Form-factor median is subtracted when supplied; the rest of the measurement model is incomplete (G01). |
| Adaptive IRT probe / p. 5 | Adaptive, not IRT | Server-owned items and marking; difficulty search is explicitly not IRT (G03). |
| BKT + deep AKT / pp. 5-6 | Heuristic substitute | Concept/reading tracks exist, but neither named model is implemented as described (G04-G05). |
| Knowledge graph / p. 6 | Logic present; data gap | Ordering, prerequisite checks and next-work selection exist; current snapshot has no prerequisite edges (G06). |
| Scaffold fading / pp. 6-7 | Implemented, incomplete | Four levels, fading and support restoration exist; assisted success can still fade support (G07). |
| Affective response / p. 7 | Cross-layer, unverified | Local helper exists; retired anxiety remains active there, and panel/action wiring needs verification (X1). |
| UDL / pp. 7-8 | Implemented backend | Three evidence types over five distinct completed lessons; session delivery settings, not a user accommodation label. |
| Module boundaries / p. 8 | Implemented backend | Six-plus segments get named delivery modules; boundary screens remain a frontend check. |
| Calculation co-construction / pp. 8-9 | Payloads and marking present | No generated pictures; read-aloud and scaffold rendering need browser verification. Marking metadata gap: G10. |
| Simplify / Expand / Slower / p. 9 | Delivery contracts present | Pending rewrites withheld; full semantic preservation is not guaranteed (G09). |
| Daily warm-up / p. 5 | Completion is tracked | Rotation and idempotent daily receipt exist; response receipt alone does not recalibrate (G02). |
| Spaced retrieval / pp. 10-11 | Scheduler implemented | Due dates, recall outcomes and daily refresh exist; durable outcome linkage needs hardening (G11). |
| No learning styles / pp. 2, 8 | Implemented in reviewed paths | Preference columns absent in profile/history; adaptation selects available delivery, not VARK matching. |
| No scores or verdicts / pp. 5, 12 | UI not certified | Some student-authorised API shapes still expose numeric internals (G12). |

BACKEND GAPS / 01-02

# Measurement must change delivery

## G01 | Baseline results do not yet implement the described measurements

**P1 / Backend + measurement specification**

**Document expects:** Reverse/complex span, flanker or Stroop interference, reading and ANS produce distinct, meaningful delivery constraints (pp. 3-5).

**Current behaviour:** Trial reduction calculates generic accuracy and latency buckets. The profile builder does not consume ANS accuracy/ratio or flanker congruency into their promised distinct estimates. Span length and complex-span interference are not explicit trial fields. Motor correction exists, but this does not make the rest of the reducer a validated measurement model. For device-generated tasks, the server can still trust the supplied correct flag.

**Fix:** Version the task/result contract: include stimulus identity, sequence length/recall, distractor condition, dot quantities/ratio and reading mode. Derive separate measured values, uncertainty and missingness; use server-verifiable task keys where practical. Define age-band and form-factor transforms with the learning-science owner. Preserve safe defaults explicitly as defaults, not measurements.

**Close when:** Changing ANS or flanker evidence changes the relevant measured setting; a skipped task produces no invented measure; identical decisions with different motor baselines yield the intended corrected comparison. Document which reading tasks are silent versus oral: no oral-fluency scoring path was established here.

**Evidence:** Source pp. 3-5; E01-E02. Controlled reducer check: opposite flanker results still produced the 12-minute default; an ANS trial produced no ANS profile field.

## G02 | A warm-up receipt is not a recalibration transaction

**P1 / Backend contract + frontend integration**

**Document expects:** The 45-second opening activity recalibrates one dimension, rather than merely recording attendance (p. 5).

**Current behaviour:** The warm-up response route marks a domain answer and stores today's receipt in preferences. It does not call the baseline reducer, revise subject placement, or update engine configuration. Device tasks could submit trials separately, but the receipt endpoint does not require or link that submission. Domain-day selection also ranges over the school bank rather than explicitly the child's enrolled subjects.

**Fix:** Link a served run ID to its task results; atomically reduce valid results and store completion once. Restrict subject selection to the enrolled curriculum. Leave unrelated dimensions unchanged; retain a clear unavailable/no-measurement outcome.

**Close when:** A measured warm-up changes only its intended parameter once; retrying it does not double-update; a receipt without valid task results cannot masquerade as a measured recalibration.

**Evidence:** Source p. 5; E03. The separate trials route can update a profile, but its use by the deployed frontend was not verified.

BACKEND GAPS / 03-05

# Named models versus current models

## G03 | The adaptive probe is not IRT

**P2 / Implement the model or correct the claim**

**Document expects:** Adaptive Item Response Theory estimates subject ability and selects informative questions (p. 5).

**Current behaviour:** selection.py explicitly says it is not IRT: correct/incorrect moves a difficulty estimate by a halving step. The bank has adaptive behaviour but no demonstrated IRT likelihood/posterior or information-based selection.

**Fix:** Either describe v1 as an adaptive difficulty-search probe, or implement a locally computed IRT estimator with item calibration, uncertainty, exposure controls and a defensible stopping rule. The latter requires suitable bank data, not just a model-name rename.

**Close when:** Documentation matches the algorithm; if IRT is retained, known synthetic ability/item cases recover the expected posterior and uncertainty, followed by cohort calibration.

**Evidence:** Source p. 5; E04. This is not evidence that the present probe is useless; it is a precise algorithm mismatch.

## G04 | Four-parameter BKT is not implemented

**P2 / Knowledge-tracing model alignment**

**Document expects:** A Bayesian update using prior, transition, guess and slip stops lucky picks becoming mastery (pp. 5-6).

**Current behaviour:** The engine uses fixed exponential increases/decreases. Guess probability is stored but not used in the update; there is no separate four-parameter BKT posterior and transition step. Slip scales the positive rate, which is not the stated Bayesian model.

**Fix:** Add a versioned local BKT update with explicit priors and item-dependent guess/slip assumptions; keep model provenance and shadow comparisons before migrating existing states. Alternatively, remove the BKT claim for v1.

**Close when:** Analytical BKT cases match, changing guess probability changes the posterior, and aided or rapid answers remain excluded from unaided-mastery evidence.

**Evidence:** Source pp. 5-6; E05. Controlled check: guess probabilities 0.2 and 0.9 both updated concept 0.4 to 0.4864.

## G05 | Deep AKT and learned reading attribution are absent

**P2 / Model development, not a small wiring fix**

**Document expects:** A deep attentive model learns relevance from interaction history and separates reading noise from concept knowledge (p. 6).

**Current behaviour:** HybridAktMasteryEngine is explicitly a pragmatic v1 rule-based substitute. Reading attribution uses fixed reading/text-density thresholds. The lesson-attempt runtime passes related_mastery={}, so its inter-concept transfer receives no related history on that path.

**Fix:** Immediately use an accurate v1 name and wire validated related-concept evidence if retaining the heuristic. A genuine AKT track needs a trained/versioned model, representative lawful data, evaluation and fallback; run it in shadow mode first.

**Close when:** Reading-confounded and concept-confounded cases are separated on held-out data; the live attempt path supplies the intended history; claims name the actually deployed model.

**Evidence:** Source p. 6; E05-E06. No model training or efficacy was established by this audit.

BACKEND GAPS / 06-07

# What comes next, and how much help

## G06 | Prerequisite logic has no populated graph to enforce

**P1 / Curriculum data + backend publishing validation**

**Document expects:** The subject probe seeds entry into a real prerequisite knowledge graph, and subsequent work follows it (p. 6).

**Current behaviour:** Next-work selection and prerequisite refusal are implemented. The current database snapshot contains 335 concepts, none with a prerequisite list. Of 402 segments, 192 lack a direct concept_id. Some can still resolve a concept through calculation/checkpoint metadata, so 192 is not a proven count of untracked sections. The absence of all graph edges means the data cannot currently demonstrate the promised prerequisite walk.

**Fix:** Build and review a school/subject curriculum graph, link every assessed question and calculation to its canonical concept, backfill mappings with a review queue, and add graph-coverage checks before lesson publication. Do not invent dependencies from title similarity alone.

**Close when:** A real approved A-to-B prerequisite blocks B until A is demonstrated, then permits it; duplicate concept names are reconciled; unmapped assessed content is explicitly identified instead of silently skipped.

**Evidence:** Source pp. 5-6; E06-E07 and read-only database aggregates. Snapshot includes test/demo content and is not a curriculum-quality judgement on every row.

## G07 | Scaffold fading can mistake assisted success for independence

**P1 / Backend state consistency**

**Document expects:** Support is contingent on demonstrated competence and fades without penalising a child who still needs it (pp. 6-7).

**Current behaviour:** Four levels, three-success fading, support restoration and logs exist. However, three correct attempts with two hints each still fade full support to partial support. The engine only rejects an increase in hint use, not continued dependence. Runtime state starts at full support independently of the displayed baseline starting level, and supplies no expected response time for its timing guard.

**Fix:** Use one authoritative session/concept support state initialised from delivery configuration; derive problem-level unaided success from marked attempts and hints across all steps. Do not fade on persistent hint dependence. Use response timing as context rather than punishing methodical correct work; preserve calm support restoration.

**Close when:** Three qualifying independent successes fade once; three hint-dependent successes do not; a struggle restores support and the next payload, dots and audit log agree.

**Evidence:** Source pp. 6-7; E06, E08-E09. Controlled check reproduced the hint-dependent fade; this is an implementation issue, not just a claim mismatch.

BACKEND GAPS / 08-09

# Delivery must follow the settings

## G08 | Some inferred pacing settings have no enforcing consumer

**P1 / Backend delivery + content coverage**

**Document expects:** Baseline capacity changes how much material is held at once, pace and pause timing; Slower delivers smaller pieces (pp. 4, 8-10).

**Current behaviour:** The backend emits segmentWordTarget, attentionWindowMinutes and comprehensionCheckInterval. Reading chunks are stable prose units with a fixed 80-word bound, not derived from those targets. The pacing fields are documented as server inference values, not client-applied controls; no adaptive chunk-sizing or check-gap enforcement consumer was found. Existing module boundaries and accommodation flags are real, but they do not by themselves implement every promised pacing effect.

**Fix:** Define a delivery adapter that maps measured settings to bounded chunk groups, pause recommendations and available check placement. Preserve stable source offsets and annotation IDs. Validate authored check coverage; flag shortages for a teacher rather than fabricating questions in the browser.

**Close when:** Two deliberately different valid configurations yield predictably different bounded delivery without losing text or annotations; a check-gap shortage is visible before publication; Slower is reversible and changes grouping rather than curriculum.

**Evidence:** Source pp. 4, 8-10; E01, E09-E11. The frontend's actual Slower rendering was not re-audited or changed.

## G09 | Simplify does not guarantee that nothing is lost

**P1 / Backend content-fidelity validation + teacher review**

**Document expects:** Simplify preserves all teaching content; Expand adds explanation without adding curriculum topics (p. 9).

**Current behaviour:** The validator preserves figures and detected equations and checks length. It does not prove coverage of non-numeric facts, definitions, conditions or safety instructions, or prevent an expanded passage introducing an unrelated claim. A controlled example dropping chlorophyll, oxygen and a light-intensity condition still passed. Teacher approval is a real safeguard: all 740 stored depth variants were pending and none approved at the time of the snapshot.

**Fix:** Extract a teacher-reviewable proposition/constraint ledger for each source, require coverage for Simplify and scope checks for Expand, and block uncertain rewrites. Keep teacher approval mandatory. Use semantic checks as a guard, not as a claim of perfect automatic proof; retain source fallback and provenance.

**Close when:** Omitting a substantive source fact, condition or warning is rejected or flagged; teachers can compare exact source and rewrite; approved variants alone reach students. Full no-loss assurance remains a review commitment, not a numeric check.

**Evidence:** Source p. 9; E12 and depth-variant database aggregate. Pending approval is school work, not a reason to bypass review.

BACKEND GAPS / 10-11

# An answer must mean the same thing everywhere

## G10 | Calculation marking drops the choice-option context

**P1 / Backend marking contract**

**Document expects:** Generated steps measure what the child actually knows, and a wrong option must not count as success (pp. 6, 8-9).

**Current behaviour:** The response model now filters choice aliases against the correct option. But _question_for_attempt builds calculation snapshots with acceptedAnswers and no options. answer_matches only filters distractor aliases when options exist. A legacy step with an invalid distractor alias can therefore be safe in the read response yet still mark that distractor correct in the attempt path. This was reproduced with the current snapshot shape, not by submitting an intentionally wrong answer to production.

**Fix:** Use the same canonical validated step for delivery and marking. Preserve option IDs and input kind, reconcile accepted versus acceptedAnswers representations, reject distractor aliases on ingest, and backfill affected historical data with an audit trail.

**Close when:** A deliberately contaminated historical alias cannot make a distractor correct; legitimate equivalent forms still work; response display and server marking agree for the same step.

**Evidence:** Source pp. 6, 8-9; E13-E14. The previous alias fix covered delivery but not this lost-metadata path.

## G11 | Review outcomes are not durably tied to the marked evidence

**P1 / Backend review transaction and retry safety**

**Document expects:** The system brings concepts back as memory fades and updates memory from actual recall (pp. 10-11).

**Current behaviour:** A local stability/retrievability scheduler, due reads, outcome vocabulary and daily refresh exist. The review write trusts a submitted outcome (or legacy boolean), accepts reviewedAt, and has no review-attempt identity for idempotency. It is separate from marked lesson attempts. Lesson completion seeds missing schedules but skips existing ones. A repeated review write can therefore update stability again, and a completed review is not guaranteed to update the existing schedule without its separate call.

**Fix:** Link a review session and concept to stored first attempts and hint use; derive recall server-side, update the schedule once in the same transaction, and accept a stable idempotency key. Restrict backdating to a defined offline-sync policy and preserve failure/retry recovery.

**Close when:** Successful unaided recall extends stability, assisted/failed recall schedules reinforcement, retries change it once, and a dropped client callback cannot leave a completed review unscheduled.

**Evidence:** Source pp. 10-11; E15-E16. The present formula already implements spaced retrieval; switching to an officially fitted FSRS model is not required by this PDF's wording.

PRIVACY BOUNDARY + CROSS-LAYER WORK

# Keep measurements behind the experience

## G12 | Student-authorised responses still expose numeric internals

**P1 / Backend response minimisation; UI audit remains separate**

**Document expects:** No child is shown scores, results, labels or a profile; feedback concerns the concept rather than a verdict about them (pp. 5, 11-12).

**Current behaviour:** BaselineSubmitResponse returns baselineProfile and engineConfig to the student. Progress and due-review schemas contain numeric understanding/reading or memory values. This does not prove the frontend renders them, and the document does not expressly prohibit their transport. It does leave a contract path for a future integration to display the very measurements the product says remain invisible.

**Fix:** Separate learner-safe delivery DTOs from staff/internal inference DTOs. Return only controls and functional progress copy needed by the player; keep baseline estimates, confidence and scheduling internals behind explicit trusted scopes. Deprecate existing fields with the frontend owner instead of silently breaking the client.

**Close when:** A student token cannot retrieve unnecessary profile/score internals from the learning reads; the player still receives its required delivery settings; a UI check confirms no evaluative labels or numerical scores are rendered.

**Evidence:** Source pp. 5, 11-12; E02, E09, E15, E17. Classified as an exposure risk, not evidence of visible scoring today.

## X1 | The affective helper needs v1 alignment and integration proof

**Cross-layer / Not a request to move raw affective data to the server**

**What exists:** repository browser-helper code keeps raw touch data in IndexedDB, excludes busy windows and includes productive-confusion grace logic. The backend intentionally does not receive a child's raw affective stream. Keep that boundary.

**Mismatch:** anxiety is still the helper's first active inference branch, although p. 7 retires it for v1. getCurrentAffectiveState returns the state string while shouldTriggerAffectiveAdaptation checks state and cooldown, not interventionReady. A caller using only those two public helpers can discard the productive-confusion grace result. No complete state-to-hint/panel/boredom-fading action dispatcher was established in the backend repository.

**Fix and acceptance:** gate anxiety off in v1; expose a local decision including interventionReady; test that productive confusion waits, frustration offers help then a break, and boredom withdraws support without a visible label. Verify the actual frontend integration and session cleanup. Keep raw events and affect labels off backend storage. Owners: frontend + backend contract + product. Evidence: source p. 7; E18.

IMPLEMENTATION ORDER + DECISIONS

# How to close the gaps

| Order | Work package | Completion evidence |
|---|---|---|
| 1 | Fix correctness paths: G10 marking; G07 support; G11 review retry safety. | Same authored answer marks consistently; aided success does not fade; review retries update once. |
| 2 | Implement measurement reduction and warm-up linkage: G01-G02. | Dimension-specific measured results change the intended setting; missing results remain unmeasured. |
| 3 | Populate curriculum links and consume delivery settings: G06, G08. | A reviewed real prerequisite chain governs entry; different settings produce different bounded delivery. |
| 4 | Harden fidelity and learner-safe contracts: G09, G12. | Fact-loss cases are caught/reviewed; only approved rewrites flow; student reads are minimised. |
| 5 | Choose honest model scope: G03-G05. | Either implement and validate named IRT/BKT/AKT models, or publish accurate v1 descriptions. |
| 6 | Verify X1 in the deployed frontend; approve content; instrument the pilot. | Local emotion decisions respect grace/retirement; no raw stream escapes; actual student experience is checked. |

## D1 | One deliberate exception needs document sign-off

The document says skipping the domain probe is not an option (p. 5). The deployed backend now allows onboarding to resolve as **unavailable** when no eligible bank exists; one available enrolled subject otherwise satisfies initial onboarding. That avoids permanent lockout and does not fabricate placement, but it is a policy change. Product should approve wording such as: "A measured subject entry point is used when a suitable bank exists; otherwise Nevo starts conservatively and records that no placement was measured." Do not reverse the lockout fix merely to match outdated prose.

## What still needs proof, not just another endpoint

The source itself calls touch-based affect inference, dimension choice and Nigerian-school outcomes reasoned rather than demonstrated (p. 11). Keep that qualification. Model calibration, false-positive rates, skill-confound separation, retention and learning outcomes need a privacy-preserving pilot protocol. Passing unit checks is not evidence of educational efficacy or a validated psychological measurement.

## Owners and changes

Backend owns the listed contracts, transactions and delivery logic. Curriculum/learning-science owners approve measurement transforms, banks and graph edges. Teachers approve rewrites. Frontend verifies rendering, autonomy controls, hidden outcomes and local affective behaviour. No dates are promised here: the deep-model work is materially larger than the contract fixes.

REPRODUCIBLE AUDIT TRAIL

# Evidence register

Paths below are relative to /Users/teslimcodes/Developer/nevo-backend-2.0 at commit 1dcf126. Line numbers identify starting points, not complete implementations. Source requirement: the supplied 12-page Nevo_Learning_Science.md.pdf; page references throughout identify the claimed mechanism.

**E01** src/nevo/intelligence/baseline.py:11, 177
Generic baseline builder and trial reducer; attention/ANS derivation coverage.

**E02** src/nevo/api/frontend_unblockers.py:423, 494, 1979
Trial schema, result DTO, motor subtraction and baseline write.

**E03** src/nevo/api/frontend_unblockers.py:2226, 2315
Warm-up receipt, daily rotation and school-bank item selection.

**E04** src/nevo/probe/selection.py:1, 67, 112
Explicit non-IRT statement; halving-step update and item difficulty recalibration.

**E05** src/nevo/mastery/engine.py:26, 90
Heuristic update, fixed reading attribution and unused guess parameter.

**E06** src/nevo/intelligence/learning_runtime.py:223, 341
Lesson attempt bridge; empty related history; scaffold state and logs.

**E07** src/nevo/intelligence/progression.py:11
Assigned next-work selection and prerequisite handling; graph snapshot described in G06.

**E08** src/nevo/intelligence/scaffolds.py:80
Hint-dependence and response-time criteria for scaffold qualification.

**E09** src/nevo/intelligence/session_delivery.py:9; src/nevo/api/response_models.py:1302
Frozen session settings and typed renderer/inference fields.

**E10** src/nevo/content_parsing/chunks.py:6
Fixed-size, stable-offset reading boundaries.

**E11** src/nevo/content_parsing/modules.py:15
Six-plus-segment named delivery-module fallback.

**E12** src/nevo/content_parsing/depth.py:112
Length, figure and equation preservation; no full proposition coverage.

**E13** src/nevo/api/product_learning.py:587, 609
Calculation marking snapshot omits choice options.

**E14** src/nevo/content_parsing/answers.py:31, 62
Distractor alias filtering requires option metadata.

**E15** src/nevo/api/scheduler.py:74, 158
Review write accepts client outcome/legacy boolean; returns memory parameters.

**E16** src/nevo/scheduler/fsrs.py:1; src/nevo/intelligence/learning_runtime.py:30
Local memory formula; completion only seeds missing schedules.

**E17** src/nevo/api/response_models.py:1459
Progress response numeric concept/reading fields.

**E18** frontend/affective/affective_inference.js:300, 338, 380
Active anxiety; cooldown-only public trigger; state-only public read.

## Additional snapshot and verification

402 segments; 335 concepts; zero concepts with prerequisite lists; 740 pending depth variants, zero approved; 11 calculation segments, three without a scaffold object. Missing scaffold objects may be deliberate fallback or may use legacy manipulative data: the count alone is not proof that all three are broken. These are current database aggregates, not an exhaustive tenant-by-tenant content audit.

54 existing checks passed across mastery, scheduling, scaffolds, accommodations, calculation contracts, adaptation regressions and affective contract checks. Controlled checks reproduced the unused-guess effect, incomplete baseline reduction, hint-assisted fading, semantic omission acceptance and calculation alias-context risk. No production lesson answers were fabricated.